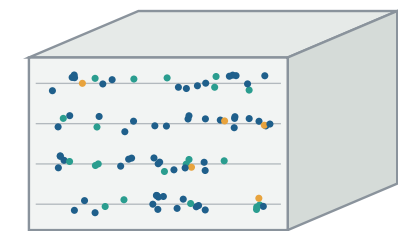


Fig. 1 MICrONS: from one imaged cortical volume to a label-free per-neuron decoding task

a one mouse, 1 mm³ of cortex, 13 scans

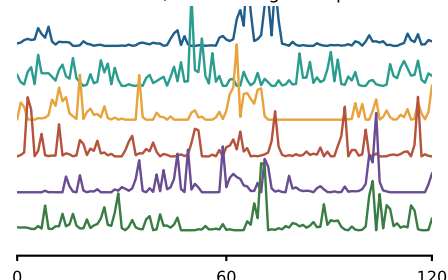
12,894 neurons, co-registered to the EM volume



● V1 ● RL ● AL ● LM

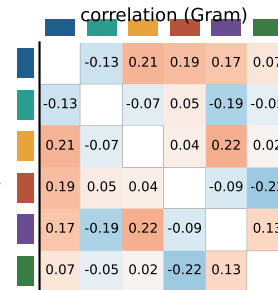
b responses to a shared stimulus → correlations

six neurons, trial-averaged response



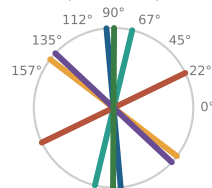
stimulus bin (oracle movie clips)

full matrix 12,894 × 12,894; the decoder sees only such matrices, never the stimulus

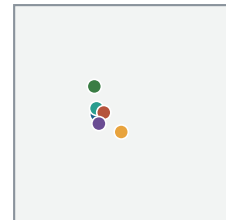


c contents (training targets only)

preferred orientation
(8 classes)

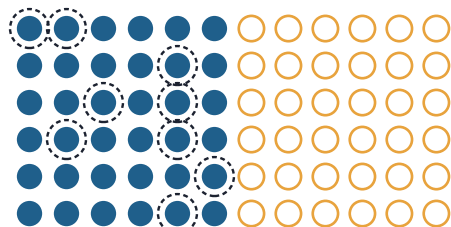


receptive-field centre
(screen)



5,287 neurons with orientation (gOSI ≥ 0.25) · 11,326 with RF

d protocol: halve the neurons, sample a population inside one half, decode every neuron of the population from its row of the Gram

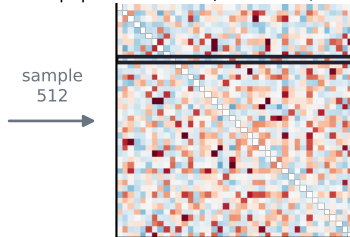


training half

test half

random split of the 12,894 neurons

population Gram (512 × 512, 40 shown)



one row = one neuron's token

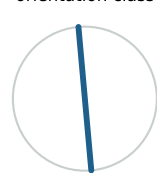
sample
512

transformer

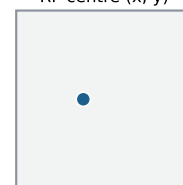
rows as tokens,
attention over the population

per-neuron prediction, scored on the test half

orientation class



RF centre (x, y)



trained on labelled training neurons;
epoch chosen on a held-out slice of the training half

no labels, no reference population in the input;
same brain, single-animal populations: the 'within' cell of Fig. 4